



Spatial patterns with memory: tree regeneration after stand-replacing disturbance in *Picea abies* mountain forests

Jan Wild, Martin Kopecký, Miroslav Svoboda, Jitka Zenáhlíková, Magda Edwards-Jonášová & Tomáš Herben

Keywords

Bark beetle outbreak; Biological legacies; Cox point processes; *Ips typographus*; Matérn process; Norway spruce; Pair-correlation function; Regeneration; Thomas process; Tree spatial pattern; Tree wells

Abbreviations

$g(r)$ = pair correlation function; $G(r)$ = nearest-neighbour distribution function; GoF = global goodness-of-fit test; CSR = complete spatial randomness

Nomenclature

Kubát et al. (2002)

Received 2 September 2013

Accepted 13 March 2014

Co-ordinating Editor: Vigdis Vandvik

Wild, J. (Corresponding author,
jan.wild@ibot.cas.cz),

Kopecký, M. (ma.kopecky@gmail.com) &

Herben, T. (herben@site.cas.cz): Institute of Botany, Academy of Sciences of the Czech Republic, CZ-252 43, Průhonice, Czech Republic

Wild, J. Faculty of Environmental Sciences, Czech University of Life Sciences Prague, Kamýcká 129, Praha 6-Suchbát, 16521, Czech Republic

Svoboda, M. (svobodam@fld.czu.cz) &

Zenáhlíková, J.

(jitka.zenahlkova@centrum.cz): Faculty of Forestry and Wood Sciences, Czech University of Life Sciences Prague, Kamýcká 129, Praha 6-Suchbát, 16521, Czech Republic

Edwards-Jonášová, M.

(edwards.m@czechglobe.cz): Global Change Research Centre, Academy of Sciences of the Czech Republic, v. v. i., Na Sádkách 7, 370 05 České Budějovice

Herben, T. (herben@site.cas.cz): Department of Botany, Faculty of Science, Charles University in Prague, Benátská 2, CZ-128 01, Praha 2, Czech Republic

Abstract

Questions: What are the spatial patterns of adult trees and recruits in natural *Picea abies* forests? How are these patterns related to each other? Does the relationship differ before and after stand-replacing disturbance? What are the ecological processes behind these spatial patterns?

Location: Mountain *Picea abies* forests in the Šumava Mts. (Bohemian Forest), Czech Republic, affected by a high-severity outbreak of bark beetle (*Ips typographus*) that caused large-scale dieback of the forest canopy.

Methods: We measured the spatial coordinates and heights of all recruits and the coordinates and DBH of all adult trees in nine plots across a wide range of recruit densities. We distinguished pre- and post-disturbance recruits, trees killed by the disturbance and trees already dead before it. To analyse spatial relationship among these groups, we used univariate and bivariate pair-correlation functions. To provide further insight into the mechanisms behind the observed patterns, we fitted Thomas and Matérn point processes to the observed data.

Results: Recruits formed tight clusters (2–9 m), whereas trees were distributed randomly or weakly clustered at short distances (1–2 m). Both pre-disturbance and post-disturbance recruits were highly clustered (cluster radii < 2 m) around trees before and after the stand-replacing disturbance. This fine-scale pattern was likely driven by a combination of: (1) seed accumulation in tree wells during winter; (2) nurse effects of tree trunks extending the vegetation period, suppressing competitive vegetation and enhancing nutrient supply from decomposed litter; and (3) suitable seedbeds on some decaying wood. The Thomas point process fitted the observed pattern of decreasing recruit density with increasing distance from mature trees better than the Matérn process.

Conclusions: Tree spatial pattern in mountain *P. abies* forests showed high resilience to stand-replacing disturbance. After a self-thinning of recruits tightly clustered around parental trees, their spatial pattern will mirror the pattern of trees that formed the stand before the disturbance. This ‘memory’ of tree spatial patterns is an important biological legacy and should be viewed as a fundamental property of natural *P. abies* forests.

Introduction

The link between spatial pattern and ecological process has been of great interest since Watt's landmark paper (Watt 1947). Important challenges remain in disentangling causal relationships between pattern and process, and the ability to infer underlying biological processes from observed patterns has been questioned (Lepš 1990). Indeed, such inference is complicated by the fact that any given observed spatial pattern is the outcome of combinations of various past events operating over different temporal and spatial scales (Lepš 1990; Dale 2000). This is especially true in forests, where processes operate over long time scales, and tree life stages differ strongly in their responses to ecological variation (Dovčiak et al. 2001; Moravie & Robert 2003). Although the effects of process on pattern has been the focus of much research, the effects that spatial patterns themselves – by influencing biological processes – can have on the spatial arrangement of successive forest stands is only rarely addressed, but see Stoll & Bergius (2005) for experimental and Murrell (2009) for theoretical approaches to this question.

In temperate forests, fine-scale tree spatial pattern is largely related to small-scale disturbances creating stand heterogeneity (Beatty & Stone 1986; Canham 1989; Diaci et al. 2005). At the landscape level, spatial heterogeneity of natural forests is thought to be maintained by large-scale disturbances such as windstorms, fires and insect outbreaks (Foster et al. 1998; Turner 2010). However, large-scale disturbances may also affect fine-scale spatial patterns of individual trees within the disturbed landscape. Because large-scale disturbances are likely to kill most mature trees, future forest development is largely determined by survival of already established recruits and new post-disturbance recruitment (Svoboda et al. 2012). Both of these processes are influenced by biological legacies – i.e. remnants of pre-disturbance forest, including surviving recruits and adult trees, dead wood and microsite heterogeneity (Swanson et al. 2011). Surprisingly little attention has been paid to the influence of their fine-scale spatial structure on post-disturbance forest development (Svoboda et al. 2010). Frelich & Reich (1995, 1999) observed that neighbourhood effects mediated by canopy trees can affect the replacement probability of tree species composition after stand-replacing disturbance, however, disturbance-related development of fine-scale spatial patterns remains unexplored. Structural complexity of early successional stages is thought to be a precursor of complexity in later stages (Donato et al. 2012), and this stresses the importance of early successional stages for forest development (Swanson et al. 2011).

Different first- and second-order statistics are used in forestry and plant ecology to describe tree spatial patterns (Haase 1995; Stoyan & Penttinen 2000; Perry et al. 2006). Ripley's *K*-function is most commonly used to distinguish random, aggregated or regular spatial patterns at different scales, however it is often criticized for its cumulative character, and non-cumulative second-order statistics, such as the pair correlation function, are preferred (Stoyan & Stoyan 1994; Wiegand & Moloney 2004; Perry et al. 2006). The advantage of these statistics is their natural interpretation as the normalized average density of individuals surrounding the focal plant at a given distance, thus constituting the 'plant's eye view' (Turkington & Harper 1979; Law et al. 2009). No single statistic can directly link spatial pattern to the underlying ecological processes, but a combination of several spatial statistics with mathematical pattern modelling through general point processes may provide greater insight on the relationship (Møller & Waagepetersen 2004; Illian et al. 2008; McIntire & Fajardo 2009). The parameters of these pure mathematical processes, such as number of clusters or cluster size, are biologically interpretable and have been used to interpret ecological drivers of observed pattern (Plotkin et al. 2002; Wiegand et al. 2007, 2009; Eichhorn 2010).

In this study, we take advantage of an exceptional situation, in which large areas of mountain Norway spruce (*Picea abies*) forest were affected by a bark beetle outbreak that was not subjected to human intervention. This allowed us to study tree spatial patterns after stand-replacing disturbances in montane Norway spruce forests. Specifically, we sought to determine:

1. What are the spatial patterns of: (1) trees that were killed by the disturbance; (2) trees that were already dead at the time of the disturbance; (3) pre-disturbance recruits; and (4) post-disturbance recruits?
2. How are these patterns related to each other, and does the relationship change after stand-replacing disturbance?
3. What are the ecological processes behind these spatial patterns?

Methods

Study area

Study plots were located in the core zone of the Bohemian Forest (Šumava National Park), which lies on the border between the Czech Republic and Germany (Fig. 1). The relief of this area comprises mountain plain with mean elevation 1100 m.a.s.l. The area is one of the largest forest complexes in Central Europe and has been protected as a national park since 1970 in Germany and 1991 in the Czech Republic.

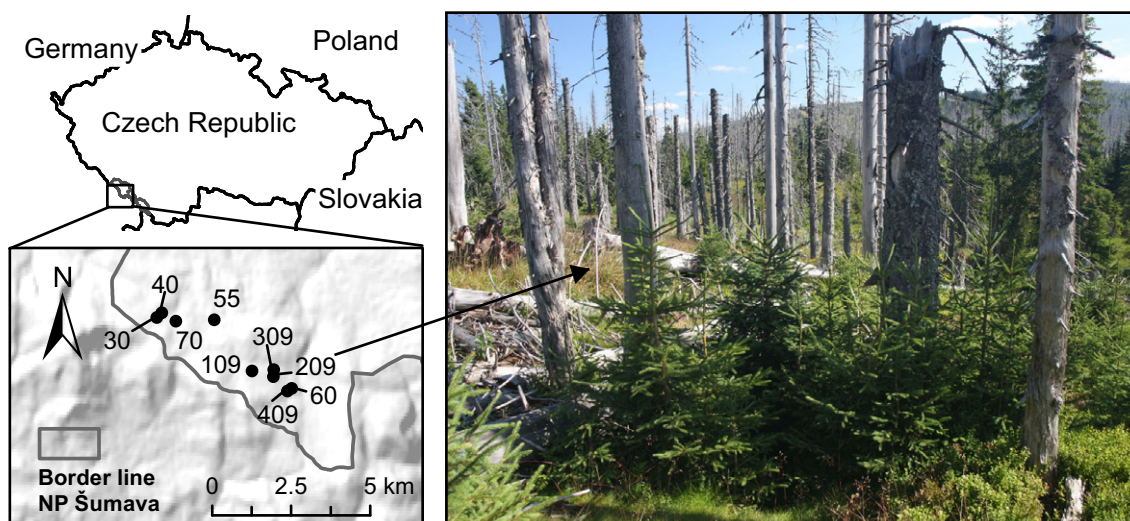


Fig. 1. Study area and overview of spatial distribution of research plots. Photograph of one study plot shows dieback of Norway spruce (*Picea abies*) tree layer after the bark beetle outbreak in the Šumava National Park, Czech Republic. Note the recruits forming tight clusters around standing snags.

Pure Norway spruce, mixed spruce–beech (*Fagus sylvatica*) forest and peatland vegetation predominate in the area (Neuhäuslová 2001; Fischer et al. 2013). The climate is relatively cold, with mean annual temperature of 4.5 °C, and humid, with annual precipitation from 1200 to 1500 mm (Sofron et al. 2001). Podsoles prevail at higher altitudes, with both podsoles and cambisols prevalent at lower altitudes (Petruš & Neuhäuslová 2001).

Forest composition and structure have been influenced by natural disturbances and management activities; windstorms followed by bark beetle outbreaks have frequently disturbed forests in this region (Brůna et al. 2013; Čada et al. 2013). The present forest represents a mosaic of logged and unlogged stands that originated after a large stand-replacing disturbance in the 1860s. The forest has a simplified vertical structure with two dominant layers – canopy and small regeneration – as is typical for European mountain spruce forests affected by large-scale disturbance (Holeksa et al. 2006; Zielonka et al. 2010). Although the forest was subject to management up to the time of national park establishment, management activities were reduced after World War II. This low-intensity management (comprising only sanitary cutting of individual uprooted or beetle-infested trees) allowed development of near-natural forest, but with a relatively low proportion of lying dead wood.

Much of this area was affected by the recent bark beetle outbreak that started in Germany after strong windstorms in 1983 and 1984 (Heurich 2001; Lausch et al. 2011). The outbreak, which peaked in the years 1996–2000, has caused the dieback of canopy trees on more than 10 000 ha (Hais et al. 2009; Lausch et al. 2011). In the

core zones of both national parks, large areas of affected forest were not subject to human intervention.

Field data collection

We established nine 40 × 40 m plots near Březník (48°58' N, 13°27' E) Czech Republic (Fig. 1). Five were established in 2008 (plot No. 30, 40, 55, 60, 70) as enlarged versions of smaller plots previously set up by M. Jonášová (Jonášová & Prach 2004), and four new plots (No. 109, 209, 309, 409) were added in 2009. All plots were located in forests with similar disturbance intensity and tree species composition. Plots were selected along the broad gradient of recruit density (Fig. 2), to be representative for the regeneration processes occurring within the area. Norway spruce was the only canopy tree species on these plots, and all canopy trees died between 1997 and 1999 due to the bark beetle (*Ips typographus*) outbreak (Fig. 1). Recruits (trees < 2.5 m in height) almost exclusively comprised Norway spruce (91–99%), with only a small amount of rowan (*Sorbus aucuparia*; 1–9%) and silver birch (*Betula pendula*; <1%). The transition between recruits and canopy trees is practically absent in these forests. No management activities were undertaken in the plots in response to the outbreak, although there had been sanitary cutting of single trees prior to bark beetle outbreak.

Positions of all live and dead trees (snags) within each plot were recorded using a laser rangefinder and an angle encoder combined with field mapping software (www.fieldmap.cz). We measured the diameter of each snag at breast height (DBH; 1.3 m) and at total tree height for snags shorter than 1.3 m.

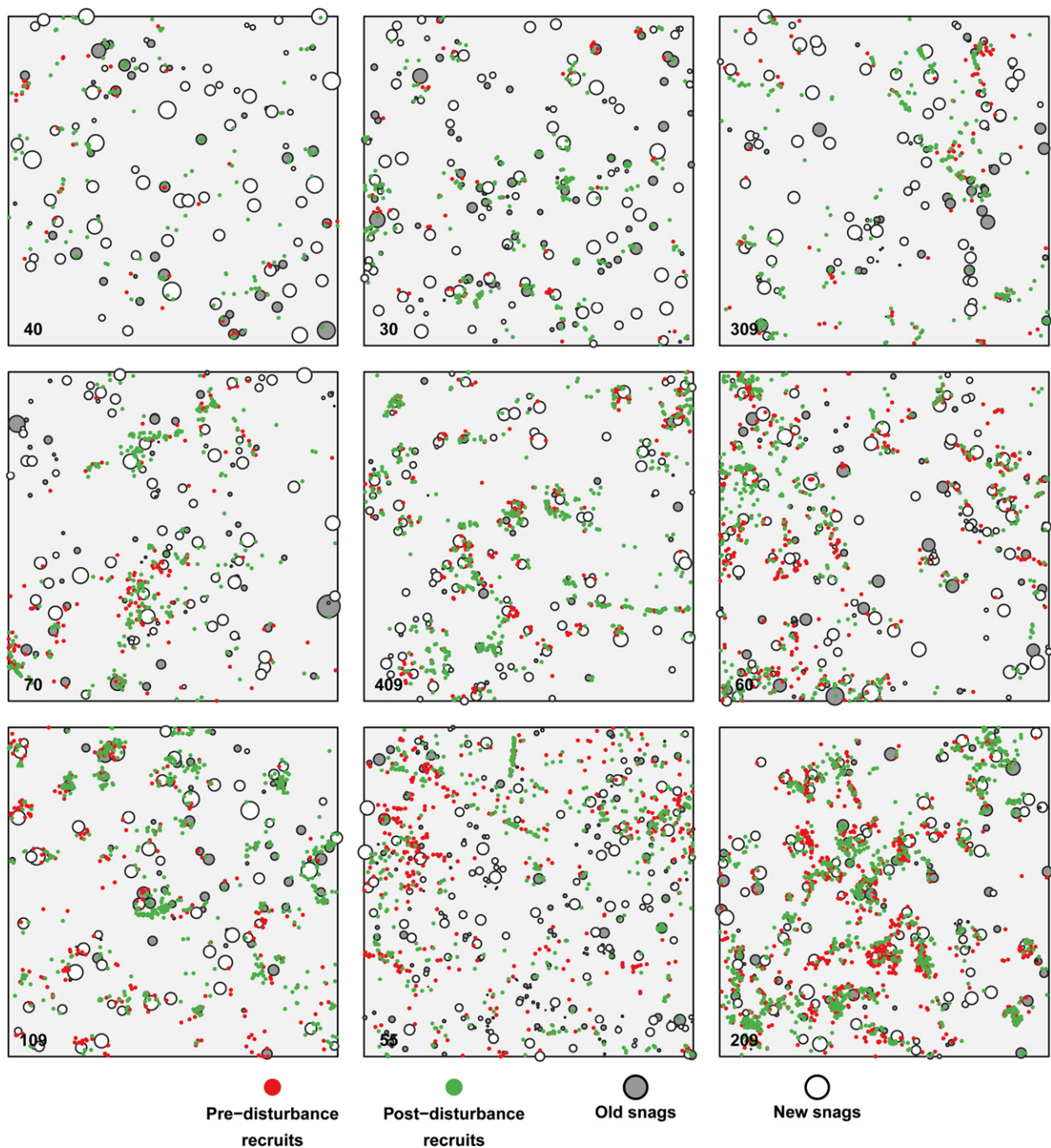


Fig. 2. Spatial patterns of pre- (red dots) and post- (green dots) disturbance recruits, and new (white) and old (grey) snags (circles with sizes proportional to stem diameter) in nine 40 × 40 m plots established in the core area of Sumava National Park, Czech Republic. Plots are ordered from low to high recruit density. Recruits formed clusters in all plots, while snags were distributed randomly.

Disturbance-related tree life stages

To analyse the spatial patterns of Norway spruce before and after the stand-replacing disturbance, we defined the following tree life stages: (1) *post-disturbance recruits*, com-

prising trees established during and after the disturbance; (2) *pre-disturbance recruits*, encompassing trees that were established, but had not yet reached maturity before the disturbance; (3) *new snags*, consisting of mature trees killed by the disturbance; and (4) *old snags*, representing mature

trees that had died before the disturbance. No living mature trees were present on any plots, and only a few sub-canopy trees occurred on several plots. The low number of sub-canopy trees precludes statistical analyses and were removed from the data set.

To discriminate between pre-disturbance and post-disturbance recruits, we determined plot-specific relationships between recruit height and age. For each plot, we estimated age by counting annual whorls and scars of 100 randomly selected individuals stratified into five height groups for equal coverage across the height distribution of recruits. To validate this approach, we counted annual rings on a subsample of all measured individuals and found a close relationship between estimated and measured age ($r = 0.75$, $N = 50$, $P < 0.001$), which is consistent with other studies (Niklasson 2002; Bače et al. 2011). We then regressed the ages of the measured individuals against their heights and estimated the approximate height of individuals established at the time of canopy dieback, as estimated from orthorectified aerial photographs. Height estimates were used as the plot-specific threshold for identification of individuals that had established before the disturbance (pre-disturbance recruits) and those that had established during and after it (post-disturbance recruits). The old and new snags were reliably distinguished by presence of bark remnants, branch thickness and wood decay stage (Fig. 1).

Univariate spatial patterns of tree life stages

To determine the fine-scale patterns of each disturbance-related tree life stage, we used a univariate pair correlation function, $g(r)$, estimated by smoothing with the Epanechnikov kernel and Ripley's isotropic edge correction (Stoyan & Stoyan 1994). Its non-cumulative character, in comparison to the traditionally used cumulative Ripley's K function, allows evaluation of spatial patterns over shorter distances (Perry et al. 2006; Law et al. 2009).

Because the 0.16-ha study plots were considered environmentally homogeneous, we tested the observed value of $g(r)$ against the null model of complete spatial randomness (CSR) to evaluate the influence of biological processes on the spatial distribution of trees. We constructed a confidence envelope at each distance, r , by removing the five highest and five lowest values of $g(r)$ of 999 random simulations of a CSR null model. Values of $g(r)$ outside the envelope indicated deviation from random spatial patterns, with $g(r) > 1$ indicating aggregation, and $g(r) < 1$ repulsion (Illian et al. 2008). Although this approach has been widely endorsed in recent work (Stoyan & Penttinen 2000; Illian et al. 2008; Law et al. 2009), it has also been criticized due to its underestimation of type I error (Loosmore & Ford 2006). Therefore, to assess general deviation from

CSR null models, we used a global goodness-of-fit (GoF) test (Diggle 2003; Loosmore & Ford 2006) calculated up to a distance of 10 m.

Spatial relationships between tree life stages

To explore spatial relationships between disturbance-related tree life stages, we used bivariate $g(r)$ (Illian et al. 2008). The interpretation of this function is similar to univariate $g(r)$, but it tested the null hypothesis that the spatial distribution of individuals belonging to different categories is independent. We tested the significance of $g(r)$ at each particular distance with a null model constructed by randomly shifting a whole group of recruit points using the toroidal boundary of the map while keeping the snags at fixed position (Goreaud & Pelissier 2003); this allows testing of the relationship between two groups of points while maintaining their intrinsic spatial patterns. We used 999 simulations and the same procedure to delineate upper and lower limits as in the univariate analysis; overall significance was also evaluated with a GoF test up to 10 m.

To test if recruits preferentially occurred around old or new snags, we counted the numbers of recruits having each of these snag types as their nearest neighbours (observed frequencies). Then we calculated standard chi-square statistics, comparing the observed frequencies against those expected from the observed proportions of old and new snags. The significance was tested by a randomization test in which snag types were randomly permuted (999 times) among all snags and the chi-square statistics were recalculated; the observed statistics were compared with the 999 randomized statistics.

Fitting general point processes to the recruits data

To enhance ecological interpretations of observed spatial patterns, we approximated recruits data by simple mathematical processes with biologically interpretable rules and parameters. We fitted two types of general Cox point cluster processes to the observed recruits data. We used the Matérn cluster process (Matérn 1960) and the Thomas process (Thomas 1949); both are biologically interpretable and have been successfully used for simulation of tree spatial patterns (Stoyan & Penttinen 2000; Pommerening & Stoyan 2006) and for uncovering ecological processes underlying observed pattern (Wiegand et al. 2009; Eichhorn 2010). Both processes generate so-called parent points as centres of clusters using a Poisson process with given intensity κ (defined by the number of centres $\bar{c}$ per area), then, a random number of offspring points, based on a Poisson distribution with mean μ , is assigned to each cluster. The points in each cluster of radius R are then distributed either randomly (Matérn cluster process) or at dis-

tances from the cluster centre according a bivariate Gaussian distribution with given standard deviation σ (Thomas process) (Møller & Waagepetersen 2004; Illian et al. 2008). Both processes generate clustered patterns, however, the closer fit of the Matérn process would indicate the microsite patches within which recruits are distributed randomly. In contrast, closer fit of the Thomas process would indicate that there is a central point in the cluster from which the density of recruits decreases.

To fit these point processes to the observed data, we used the minimum-contrast method, which searches for the closest match between the theoretical $g(r)$ of the process with unknown parameters and the nonparametric estimate of $g(r)$ from the observed data (Diggle 2003; Møller & Waagepetersen 2004). The integrated squared difference between the fourth roots of the theoretical and observed statistics was minimized using the *mincontrast* function of R-package *spatstat* (Baddeley & Turner 2005).

We tested the deviation of the observed pattern from the pattern generated by the particular cluster process using the simulation envelopes based on $g(r)$ of the process. However, the fit of $g(r)$ is not sufficient evidence that a particular cluster process describes an observed pattern well (Wiegand et al. 2009). Therefore, we used the nearest-neighbour distribution function $G(r)$ calculated without edge correction, showing the probability of a recruit having its nearest neighbour at distance r (Diggle 2003; Illian et al. 2008). The overall deviance from the particular cluster process was tested for GoF up to 10 m for $g(r)$ and up to ca. 2–4 m for $G(r)$, depending on the maximal nearest-neighbour distance of recruits on particular plots.

To gain further insight into mechanisms generating observed patterns, we have decomposed clustered and isolated components of recruit pattern following Wiegand et al. (2009). We defined a cluster component as recruits with the nearest neighbour at a distance shorter than 1.5 m, and isolated component as recruits with no neighbour within 1.5 m distance. Then we fit the Thomas and Matérn point processes only to the cluster component pattern and compared them with the processes fitted to all recruit data. We used univariate $g(r)$ to evaluate the spatial pattern of the isolated component.

All analyses were performed using R 2.15.3 (R Foundation for Statistical Computing, Vienna, AT) with *spatstat* R-package (Baddeley & Turner 2005).

Results

Basic stand characteristics

The number of spruce recruits varied between 1175 and 12 388 individuals ha^{-1} , with mean height of 71 cm (± 43 SD). Post-disturbance recruits accounted for 54–80% of all recruits on all plots. The number of new snags varied from

331 to 750 individuals ha^{-1} , with a mean DBH of 34 ± 13 cm. Old snags occurred at frequencies of 238–1006 individuals ha^{-1} , with a mean DBH of 22 ± 12 cm. Both old and new snags exhibited unimodal bell-shaped diameter distributions (Fig. 3). A detailed overview of the data is presented in Table 1, and Fig. 2 represents the spatial distribution of all mapped individuals.

Univariate patterns of tree life stages

We found significant aggregation of recruits on all plots for distances of 2–9 m. Both pre- and post-disturbance recruits exhibited similar aggregated spatial patterns (Fig. 4). Old and new snags were either randomly distributed or weakly aggregated at short distances to 2 m (Fig. 4); similar results were achieved when all snags were analysed together (not displayed).

Spatial relationships between tree life stages

We found aggregation of recruits around old and new snags up to a distance of 2 m, but new snags exhibited a lower intensity (Fig. 5). New snags displayed an offset of clustering from 0 m due to the presence of trunks, making the occurrence of recruits at that distance impossible. In contrast, some old snags were suitable seedbeds and clustering was evident from 0 m. Similar to the univariate analysis, we did not find any differences in bivariate patterns along the gradient of recruit density (Fig. 5). Separate analyses of the two types of recruits indicated higher aggregation of post-disturbance recruits around snags, particularly old snags. Pre-disturbance recruits were also clustered around snags, but the clustering tendency was lower, and new snags seemed to be preferred (Fig. 5).

The comparison of clustering around new and old snags, expressed as number of recruits occurring nearby, did not show any clear differences (Appendix S1). Significantly more recruits were clustered around old snags in two plots, with no significant difference evident in seven plots. Similar results were observed when we analysed only recruits within 1 m of snags with DBH > 15 cm. For pre-disturbance and post-disturbance recruits, significantly higher numbers of recruits were clustered around old snags on only one and three plots, respectively. Similar results were observed when the restrictions on distance of recruits from tree trunks and DBH of snags were applied (see Appendix S1).

Fitting general point processes to the recruits data

Both Cox processes fit the observed recruits data reasonably well. The pair correlation functions for all plots were within the simulated envelopes (Appendix S2). The Thomas process fit the observed pattern closer, yielding a

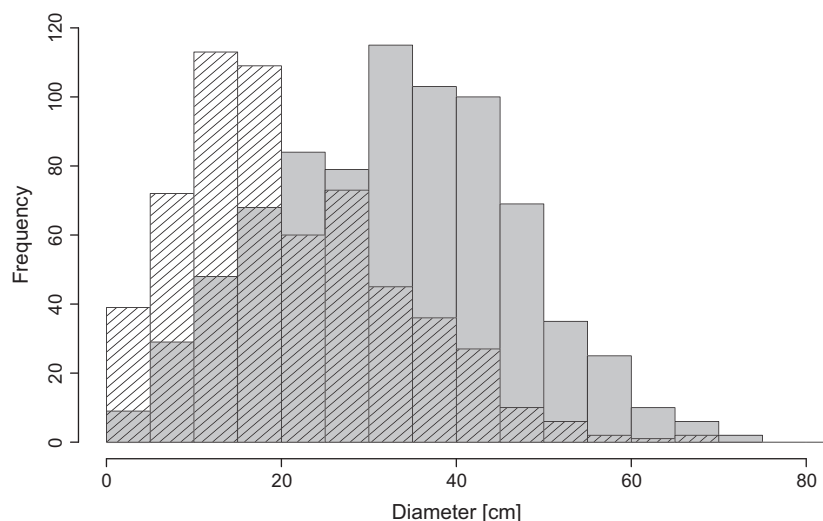


Fig. 3. Diameter distribution of old (hatched) and new snags (grey) among all plots. Old snags died before bark beetle outbreak mostly due to natural stand self-thinning. New snags represent trees killed directly by the bark beetle outbreak.

Table 1. Basic characteristics of forest stands on study plots. Numbers are rounded to reasonable numbers of digits and are given per 40×40 m plot. Plots are ordered from low to high recruit density.

Plot No.	# Pre Recruits	Mean Height [cm] (SD)	# Post Recruits	Mean Height [cm] (SD)	# Old Snags	Mean DBH [cm] (SD)	# New Snags	Mean DBH [cm] (SD)
40	49	144 (55.3)	138	57 (21.5)	38	32.1 (13.0)	68	40.6 (15.9)
30	71	125 (29.0)	276	57 (21.9)	86	22.5 (11.3)	88	34.6 (11.8)
309	112	137 (45.4)	298	55 (22.8)	38	27.1 (12.9)	78	33.8 (14.1)
70	160	80 (23.6)	305	32 (11.6)	55	21.9 (16.1)	77	35.5 (12.4)
409	230	167 (45.3)	540	71 (28.4)	29	19.0 (9.1)	102	30.7 (11.1)
60	373	81 (26.5)	442	37 (11)	59	25.4 (14.3)	110	32.3 (13.9)
109	245	130 (42.6)	587	47 (20.2)	54	27.9 (11.0)	53	43.5 (12.9)
55	422	89 (37.4)	503	35 (11)	161	14.5 (9.6)	120	27.3 (10.6)
209	867	110 (29.5)	1114	49 (17.7)	76	25.7 (12.0)	86	33.7 (13.4)

non-significant value in the GoF test (i.e. no deviation from the null model) for eight plots compared to only five plots for the Matérn cluster process. When we fitted Cox processes separately to pre-disturbance and post-disturbance recruits, the results were qualitatively similar. Although the pair correlation function fit the observed pattern well, the nearest neighbour distribution function, $G(r)$, indicated some deviation from the modelled pattern (Appendix S2). This was supported by significant results from GoF tests for all plots and both cluster processes. In almost all cases, the observed $G(r)$ was slightly above the expected $G(r)$ at small distances up to 0.5 m, thus indicating more intense clustering at shorter distances than predicted by the cluster processes. For distances > 0.5 m, the observed $G(r)$ appeared below the expected curve; this suggests the occurrence of a random process operating at larger distances and the superposition of cluster and random processes. Therefore, we decomposed the observed pattern into cluster and isolated components (Appendix S3). The isolated component pattern formed only a small

portion of the recruits (2–13%; Table 2), but the decomposition substantially improved the fit of the Thomas process to the cluster component (Fig. 6), as indicated by the non-significant GoF test for the fitted $G(r)$ function at distance (0.5–4.0 m) for five plots. The isolated pattern itself was random. The observed pattern was best described by superposition of the Thomas process and a random pattern.

Two basic parameters of the Thomas cluster process can be interpreted biologically: (1) the intensity, κ , of the process, which refers to the number of cluster centres, and (2) the SD of the Gaussian distribution, which could serve as a proxy for cluster size, given as 1.96σ . The number of cluster centres simulated by the Thomas process for the cluster component varied between 11 and 65 for each plot (Table 2), which is much lower than the observed total number of snags (106–281) and lower than the number of snags with DBH > 20 cm (84–119). The estimated average radii of clusters varied from 1.18 to 2.02 m among all studied plots (Table 2), and roughly corresponded to the size of aggregation indicated by the bivariate $g(r)$ analysis (Fig. 5).

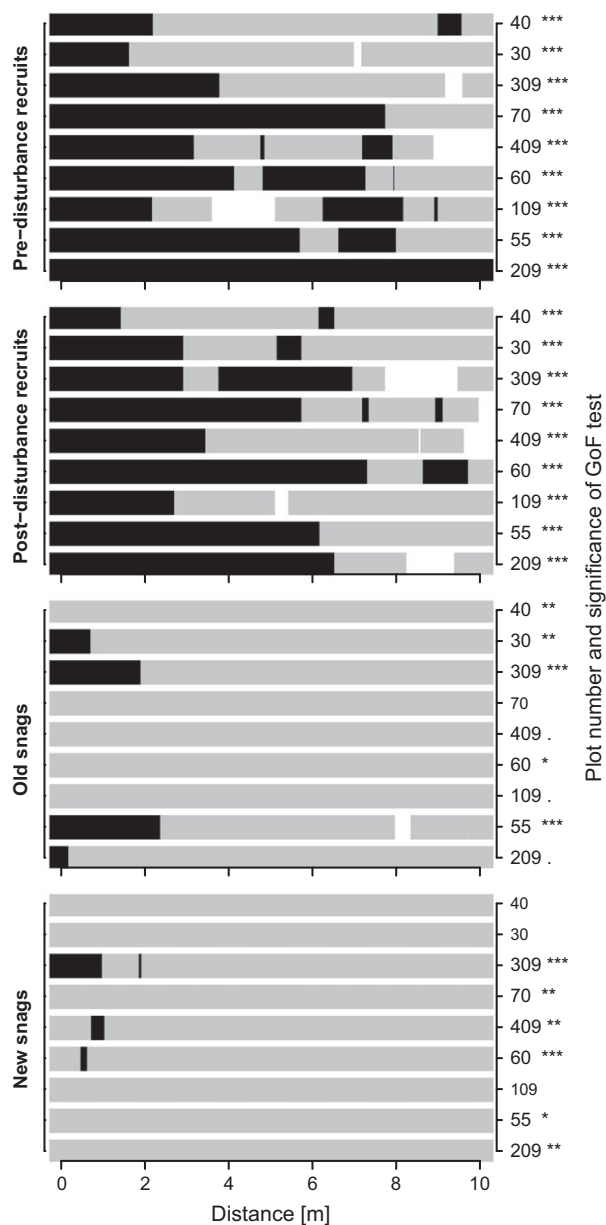


Fig. 4. Spatial patterns of different disturbance-related life stages of Norway spruce after bark beetle outbreak, evaluated with univariate pair correlation function and CSR null model. Black = clustering; grey = random pattern; white = repulsion at given lag distance. GoF test indicates probability of pattern deviation from a CSR null model; *** < 0.001, ** < 0.01, * < 0.05, . < 0.1. Plots are ordered from low to high recruit density.

Discussion

Spatial patterns of spruce recruitment

We found intensive aggregation of spruce recruits up to 9 m on all studied plots, regardless of recruit density, which could indicate the formation of recruit clusters in canopy

gaps (Canham 1989). However, we also found that recruits were strongly clustered around trees recently killed by the bark beetle outbreak, and trees that were already dead before the disturbance. The observed pattern is likely not related to gaps and may have been generated by other processes that affect seed dispersal, seedling establishment or subsequent survival. We tried to identify possible mechanisms and discuss the implications of our findings for future forest development and forest management.

Seed dispersal

New snags represent canopy trees that were still alive or had recently been killed by bark beetles at the time when pre-disturbance recruits were establishing. Limited dispersal of seeds could provide a straightforward explanation for the observed clustering around new snags. This is also supported by the peak of seed deposition close to trunk bases (0.5–2.1 m), predicted by seed trapping and inverse fitting of overlapped seed shadows of individual *Picea abies* trees (Dovčiak et al. 2008). However, pre-disturbance recruits were also clustered around old snags that would not have produced any seeds. Post-disturbance recruits were clustered around new snags, which also do not produce any seeds. Dispersal curves modelled for individual coniferous trees predict seed deposition peaks at larger distances than inverse fitting of overlapped seed shadows (Greene & Johnson 1989; Nathan et al. 2000), thus other mechanisms must be influencing the observed fine-scale clustering around tree bases.

We propose that the occurrence of tree wells (also referred to as melt holes) that form in the snow around tree trunks (Sturm 1992; Geiger 2009), especially conifers, act as a sink for seed accumulation and deposition. Tree wells are formed due to snow interception by branches, higher absorption of shortwave radiation by trunks, and the reduction in albedo due to litter accumulation on top of the snowpack (Hardy et al. 2000). Tree wells are created around live trees and snags, and may serve as effective seed traps, catching spruce seeds dispersed over crusted snow (pers. obs.). Such secondary seed dispersal is thought to be most important for small herbaceous plants (Greene & Johnson 1997) and has been only rarely reported for trees (Matlack 1989). Norway spruce disseminates the majority of its seeds in winter (Messer 1956), and our results suggest that tree wells influence Norway spruce seedling spatial patterns and are an integral part of regeneration processes in mountain spruce forests.

Establishment and survival

The close association of recruits with snags could be further influenced by microsite conditions near the base of trees

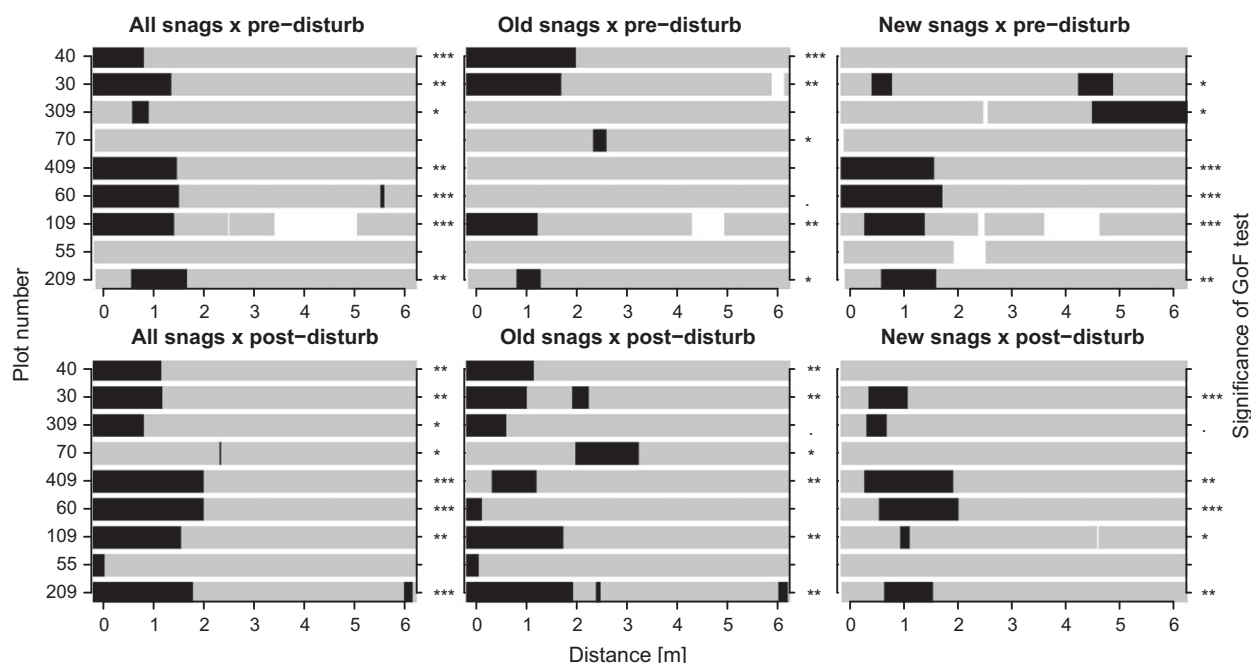


Fig. 5. Spatial relationships between different types of recruits and different types of snags evaluated using bivariate pair correlation function and null model based on toroidal shift of regeneration pattern. Black = clustering; grey = random pattern; white = repulsion at given lag distance. Significance of goodness of fit (GoF) test indicate overall deviation of observed pattern from a null CSR model; *** < 0.001, ** < 0.01, * < 0.05, · < 0.1. Plots are ordered from low to high recruit density.

Table 2. Parameters of Thomas cluster process fitted to the observed data: κ is the intensity of the point process and refers to the number of cluster centres; σ is the SD of the Gaussian distribution, which is used for distribution of random number of offspring following a Poisson distribution with mean μ . Cluster radii were calculated as $1.96 \times \text{SD}$, to contain ca. 95% of generated points. Cluster and isolated components of recruits show the numbers of points having nearest neighbour distance to other recruit lower or higher than 1.5 m, respectively. Plots are ordered from low to high recruit density.

Plot No.	No. of Centres	κ	σ	μ	Cluster Radius [m]	Cluster Component	Isolated Component
40	55.9	0.035	0.388	3.2	1.22	179	24
30	39.7	0.025	0.54	8.49	1.44	337	15
309	29.8	0.019	0.581	12.58	1.49	375	35
70	10.9	0.007	1.132	41.12	2.09	448	42
409	46	0.029	0.687	16.34	1.62	751	21
60	26.5	0.017	1.211	31.33	2.16	829	27
109	54.7	0.034	0.565	14.9	1.47	815	19
55	64.9	0.041	0.706	13.89	1.65	901	31
209	55.7	0.035	0.736	35.27	1.68	1966	16

that allow for higher seedling establishment and survival. Recruits growing closer to tree trunks experience longer growing seasons because spring snowmelt finishes earlier in tree wells due to lower snowpack and higher absorption of radiation by trunks (Pomeroy et al. 2009). This mechanism could result in clustering around both living trees and standing snags and contribute to the observed pattern.

Norway spruce recruits are highly sensitive to competition with forest floor vegetation, especially tall grasses, such as *Calamagrostis villosa*, that could inhibit seedling establishment (Diaci 2002; Holeksa 2003; Baier et al.

2005; Zielonka 2006). Bases of living trees are usually less covered with forest floor vegetation (Holeksa 2003; Kirchner et al. 2011) and represent a less competitive environment for spruce seedlings. Also, better nutrient conditions are characteristic of suitable seedbeds (Baier et al. 2005), although the effect of enhanced nutrient supply near living Norway spruce tree trunks is very weak (Koch & Matzner 1993; Seiler & Matzner 1995). However, after bark beetle outbreak, the amount of litter deposited close to trunks greatly increases (Kovářová & Vacek 2003; Bauer et al. 2008). Lower levels of competing floor vegetation and

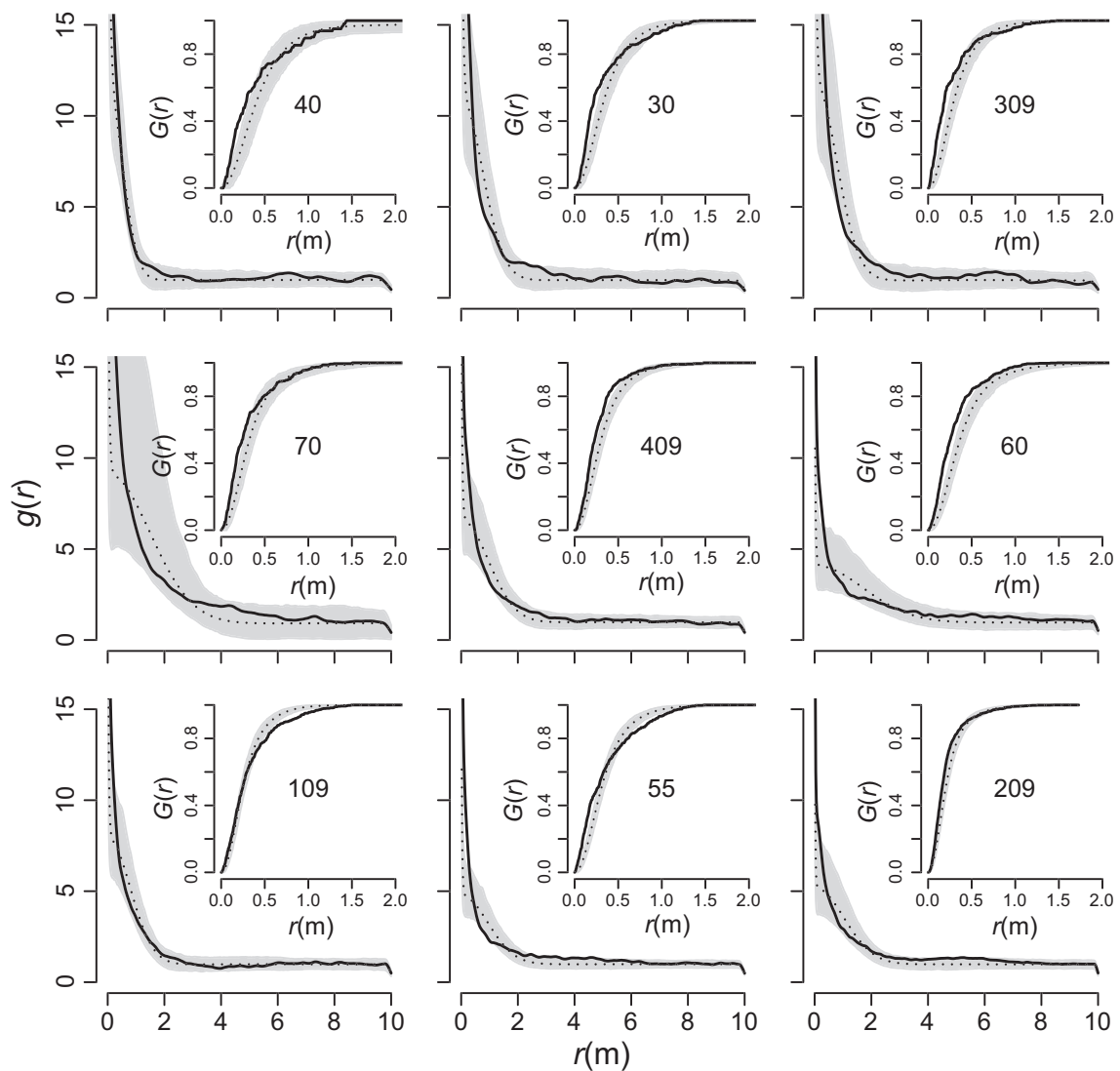


Fig. 6. Fit of observed pattern (black solid line) with Thomas cluster process (dotted line) as assessed by pair correlation function $g(r)$ and nearest neighborhood distribution function $G(r)$ (inset graphs). Envelopes (grey) are based on five highest and five lowest values from 999 simulations of CSR null model. The pattern is a superposition of random and cluster processes. Fit of cluster component of the observed pattern defined by nearest neighbour distance between recruits < 1.5 m is shown. Plots are ordered from low to high recruit density.

enhanced nutrient supply (Kovářová & Vacek 2003) at the base of trees may support post-disturbance recruits and further reinforce the observed clustered patterns.

Dead wood as suitable seedbed

The category 'old snags' was mostly comprised of stubs (dead trees missing crowns but with a substantial portion of the trunk remaining) and stumps in various states of decomposition. These likely served as seedbeds, because we found a positive relationship between old snags and recruits starting at 0 m. Decaying wood is often described

as the most suitable seedbed in natural spruce forests (Motta et al. 2002; Jonášová & Prach 2004; Lonsdale et al. 2008; Bače et al. 2012); it provides recruits with advantages through improved nutrient and moisture conditions (Bače et al. 2012), and the elevated position may help protect young seedlings from the competitive herb layer (Diaci 2002; Holeksa 2003; Baier et al. 2005; Zielonka 2006). Decaying wood is, however, not the dominant factor behind the fine-scale clustering; clusters were formed around trees living at the time of recruit establishment, and this could not be ascribed to dead wood effects.

Simulating the observed pattern

The Thomas process modelled the occurrence of points in clusters of decreasing density with increasing distance from the centre with radii of ca. 1.5–2.0 m, similar to the observed pattern. Nevertheless, the number of cluster centres simulated by the Thomas process was much lower than the number of trees observed, even after limiting the analysis to individuals with DBH > 20 cm. This discrepancy between the observed number of snags (i.e. potential cluster centres) and simulated number of clusters is likely caused by the following factors: (1) snags were distributed randomly or clustered at short distances, and they often formed small clusters of several individuals (Fig. 2). Although composed of several snags, such a small cluster still represents one potential cluster centre for regeneration. (2) Not all snags produce tree wells; snags that are in an advanced state of decomposition may be fully covered by snow in winter. (3) Short snags are not always suitable seedbeds; suitability depends on the degree of decomposition, moss cover, contact with the ground and snag diameter (Takahashi et al. 2000; Bače et al. 2012).

The nearest neighbour distribution and splitting the point process into clustered and isolated components exhibited higher observed than simulated intensity of clustering at smaller (<0.5 m) distances. This can be explained from the effects of tree wells, which formed tightly around the trunks. Old snags serving as seedbeds can also increase the clustering intensity in comparison to that predicted with the Thomas process.

The detected isolated component pattern formed only a small proportion of all recruits; this indicates that factors other than clustering around tree trunks also influenced observed spatial patterns. Besides dead wood, other microsites without competitive herb layer, such as bare ground, patches of *Sphagnum* and other mosses, are preferred by spruce seedlings (Kirchner et al. 2011; Zenáhlíková et al. 2011). They are created independently of snag positions, and the pre-existing heterogeneity of these microsites persists several years after disturbance (Wild 2001). They may explain the presence of an isolated component pattern not related to snags, however, this component is relatively minor.

Conclusion: spatial patterns with memory

Over time, tree spatial patterns develop through self-thinning from aggregated to random or even regular (Kenkel 1988; Kenkel & Hendrie 1997). We found that the fine-scale distribution of recruits in mountain spruce forest was mainly dictated by the spatial pattern of adult trees. Clusters formed by recruits were small; as clusters develop over

time, we could expect survival of few, and, finally, only one adult tree per cluster. The trees would thus mirror the spatial pattern of the prior forest stand. Tree spatial pattern itself should therefore be considered an important biological legacy governing the regeneration process of mountain spruce forests. We suggest that this memory of previous spatial patterns is driven by: (1) the accumulation of seeds related to seed trapping in tree wells during the winter; (2) nurse effects of tree trunks, consisting of extended vegetation periods in tree wells, suppressed competitive vegetation and enhanced nutrient supply from decomposed litter; and (3) suitable seedbeds on some decaying snags.

These findings have important implications for forest management after bark beetle outbreaks. Currently applied salvage logging removes all standing snags, which form tree wells, identified by us as the likely driver of the observed recruitment patterns. Salvage logging therefore directly destroys the mechanism largely responsible for the specific spatial patterns in mountain Norway spruce forests, and also damages the recruits highly concentrated around tree bases. Because of the important effect of recruit spatial pattern on the structure of future forest, spatial pattern should be recognized as an important ecosystem property, and maintained by natural processes, especially in protected areas.

Acknowledgements

We thank all the students who helped with field data collection. Jonathan Rosenthal and Robert C. Morrissy provided comments and corrections to this text. We thank the editor and two referees for their thorough reading and helpful comments. The research was supported by the Czech Science Foundation (grant No. P504/10/0843) and by the Ministry of the Environment (grant No. Sp/2d2/111/08). JW, MK and TH were further supported by the Academy of Science of the Czech Republic (long-term research development project no. RVO 67985939).

References

- Bače, R., Svoboda, M. & Janda, P. 2011. Density and height structure of seedlings in subalpine spruce forests of Central Europe: logs vs. stumps as a favourable substrate. *Silva Fennica* 45: 1065–1078.
- Bače, R., Svoboda, M., Pouska, V., Janda, P. & Červenka, J. 2012. Natural regeneration in Central-European subalpine spruce forests: which logs are suitable for seedling recruitment? *Forest Ecology and Management* 266: 254–262.
- Baddeley, A. & Turner, R. 2005. Spatstat: an R package for analyzing spatial point patterns. *Journal of Statistical Software* 12: 1–42.

- Baier, R., Meyer, J. & Göttlein, A. 2005. Regeneration niches of Norway spruce (*Picea abies* [L.] Karst.) saplings in small canopy gaps in mixed mountain forests of the Bavarian Limestone Alps. *European Journal of Forest Research* 126: 11–22.
- Bauer, M.L., Fischer, A., El Kate, H. & Mosandl, R. 2008. Verjüngungsdynamik nach großflächigem Borkenkäferbefall in den Fichtenwäldern der Hochlagen des Bayerischen Waldes. *Allgemeine Forst und Jagdzeitung* 179: 43–52.
- Beatty, S.W. & Stone, E.L. 1986. The variety of soil microsites created by tree falls. *Canadian Journal of Forest Research* 16: 539–548.
- Brůna, J., Wild, J., Svoboda, M., Heurich, M. & Müllerová, J. 2013. Impacts and underlying factors of landscape-scale, historical disturbance of mountain forest identified using archival documents. *Forest Ecology and Management* 305: 294–306.
- Čada, V., Svoboda, M. & Janda, M. 2013. Dendrochronological reconstruction of the disturbance history and past development of the mountain Norway spruce in the Bohemian Forest, central Europe. *Forest Ecology and Management* 295: 59–68.
- Canham, C.D. 1989. Different responses to gaps among shade-tolerant trees species. *Ecology* 70: 548–550.
- Dale, M.R.T. 2000. *Spatial pattern analysis in plant ecology*. Cambridge University Press, Cambridge, UK.
- Diaci, J. 2002. Regeneration dynamics in a Norway spruce plantation on a silver fir–beech forest site in the Slovenian Alps. *Forest Ecology and Management* 161: 27–38.
- Diaci, J., Pisek, R. & Boncina, A. 2005. Regeneration in experimental gaps of subalpine *Picea abies* forest in the Slovenian Alps. *European Journal of Forest Research* 124: 29–36.
- Diggle, P.J. 2003. *Statistical analysis of spatial point patterns*. Arnold, London, UK.
- Donato, D.C., Campbell, J.L. & Franklin, J.F. 2012. Multiple successional pathways and precocity in forest development: can some forests be born complex? *Journal of Vegetation Science* 23: 576–584.
- Dovčiak, M., Frelich, L.E. & Reich, P.B. 2001. Discordance in spatial patterns of white pine (*Pinus strobus*) size-classes in a patchy near-boreal forest. *Journal of Ecology* 89: 280–291.
- Dovčiak, M., Hrivnák, R., Ujházy, K. & Gömöry, D. 2008. Seed rain and environmental controls on invasion of *Picea abies* into grassland. *Plant Ecology* 194: 135–148.
- Eichhorn, M.P. 2010. Pattern reveals process: spatial organisation of a Kamchatkan stone birch forest. *Plant Ecology & Diversity* 3: 281–288.
- Fischer, H.S., Winter, S., Lohberger, E., Jehl, H. & Fischer, A. 2013. Improving transboundary maps of potential natural vegetation using statistical modeling based on environmental predictors. *Folia Geobotanica* 48: 115–135.
- Foster, D.R., Knight, D.H. & Franklin, J.F. 1998. Landscape patterns and legacies resulting from large, infrequent forest disturbances. *Ecosystems* 1: 497–510.
- Frelich, L.E. & Reich, P.B. 1995. Neighborhood effects, disturbance, and succession in forests of the western Great Lakes region. *Ecoscience* 2: 148–158.
- Frelich, L.E. & Reich, P.B. 1999. Neighborhood effects, disturbance severity, and community stability in forests. *Ecosystems* 2: 151–166.
- Geiger, R. 2009. *The climate near the ground*. Harvard University Press, Cambridge, MA, US.
- Goreaud, F. & Pelissier, R. 2003. Avoiding misinterpretation of biotic interactions with the intertype K-12-function: population independence vs. random labelling hypotheses. *Journal of Vegetation Science* 14: 681–692.
- Greene, D. & Johnson, E. 1989. A model of wind dispersal of winged or plumed seeds. *Ecology* 70: 339–347.
- Greene, D. & Johnson, E. 1997. Secondary dispersal of tree seeds on snow. *Journal of Ecology* 85: 329–340.
- Haase, P. 1995. Spatial pattern-analysis in ecology based on Ripley K-function – introduction and methods of edge correction. *Journal of Vegetation Science* 6: 575–582.
- Hais, M., Jonášová, M., Langhammer, J. & Kučera, T. 2009. Comparison of two types of forest disturbance using multi-temporal Landsat TM/ETM+ imagery and field vegetation data. *Remote Sensing of Environment* 113: 835–845.
- Hardy, J., Melloh, R. & Robinson, P. 2000. Incorporating effects of forest litter in a snow process model. *Hydrological Processes* 14: 3227–3237.
- Heurich, M. 2001. Waldentwicklung im montanen Fichtenwald nach großflächigem Buchdruckerbefall im Nationalpark Bayerischer Wald. *Waldentwicklung im Bergwald nach Windwurf und Borkenkäferbefall Nationalpark Bayerischer Wald, Wissenschaftliche Reihe* 14: 99–177.
- Holeksa, J. 2003. Relationship between field-layer vegetation and canopy openings in a Carpathian subalpine spruce forest. *Plant Ecology* 168: 57–67.
- Holeksa, J., Saniga, M., Szwagrzyk, J., Dziedzic, T., Ferenc, S. & Wodka, M. 2006. Altitudinal variability of stand structure and regeneration in the subalpine spruce forests of the Pol'ana biosphere reserve, Central Slovakia. *European Journal of Forest Research* 126: 303–313.
- Illian, J., Penttinen, A., Stoyan, H. & Stoyan, D. 2008. *Statistical analysis and modelling of spatial point patterns*. John Wiley & Sons, Chichester, UK.
- Jonášová, M. & Prach, K. 2004. Central-European mountain spruce (*Picea abies* (L.) Karst.) forests: regeneration of tree species after a bark beetle outbreak. *Ecological Engineering* 23: 15–27.
- Kenkel, N. 1988. Pattern of self-thinning in jack pine: testing the random mortality hypothesis. *Ecology* 69: 1017–1024.
- Kenkel, N. & Hendrie, M. 1997. A long-term study of *Pinus banksiana* population dynamics. *Journal of Vegetation Science* 8: 241–254.
- Kirchner, K., Kathke, S. & Bruehlheide, H. 2011. The interaction of gap age and microsite for herb layer species in a near-natural spruce forest. *Journal of Vegetation Science* 22: 85–95.

- Koch, A.S. & Matzner, E. 1993. Heterogeneity of soil and soil solution chemistry under Norway spruce (*Picea abies* Karst.) and European beech (*Fagus sylvatica* L.) as influenced by distance from the stem basis. *Plant and Soil* 151: 227–237.
- Kovářová, M. & Vacek, S. 2003. Mountain Norway spruce forests: needle supply and its nutrient content. *Journal of Forest Science* 49: 327–332.
- Kubát, K., Hrouda, L., Chrtěk, J. Jr, Kaplan, Z., Kirschner, J., Štěpánek, J. & Zázvorka, J. 2002. *Klíč ke květeně České republiky* [Key to the flora of the Czech Republic]. Academia, Praha, CZ.
- Lausch, A., Fahse, L. & Heurich, M. 2011. Factors affecting the spatio-temporal dispersion of *Ips typographus* (L.) in Bavarian Forest National Park: a long-term quantitative landscape-level analysis. *Forest Ecology and Management* 261: 233–245.
- Law, R., Illian, J., Burslem, D.F.R.P., Gratzner, G., Gunatilleke, C.V.S. & Gunatilleke, I.A.U.N. 2009. Ecological information from spatial patterns of plants: insights from point process theory. *Journal of Ecology* 97: 616–628.
- Lepš, J. 1990. Can underlying mechanisms be deduced from observed patterns? In: Krahulec, F., Agnew, A.D.Q., Agnew, S. & Willems, J.H. (eds.) *Spatial processes in plant communities*, pp. 1–11. Academia, Prague, CZ.
- Lonsdale, D., Pautasso, M. & Holdenrieder, O. 2008. Wood-decaying fungi in the forest: conservation needs and management options. *European Journal of Forest Research* 127: 1–22.
- Loosmore, N.B. & Ford, E.D. 2006. Statistical inference using the G or K point pattern spatial statistics. *Ecology* 87: 1925–1931.
- Matérn, B. 1960. Spatial variation: stochastic models and their applications to problems in forest surveys and other sampling investigations. *Meddelanden från Statens Skogsforskningsinstitut* 49: 1–144.
- Matlack, G.R. 1989. Secondary dispersal of seed across snow in *Betula lenta*, a gap-colonizing tree species. *Journal of Ecology* 77: 853–869.
- McIntire, E.J.B. & Fajardo, A. 2009. Beyond description: the active and effective way to infer processes from spatial patterns. *Ecology* 90: 46–56.
- Messer, H. 1956. Untersuchungen über das Fruchten der Fichte (*Picea abies*). In: Messer, H. (ed.). *Forstschritte des forstlichen Saatgutwesens.*, pp. 91–117. J.D. Sauerlander's Verlag, Frankfurt a. Main, DE.
- Møller, J. & Waagepetersen, R.P. 2004. *Statistical inference and simulation for spatial point processes*. Chapman & Hall/CRC, London, UK.
- Moravie, M.-A. & Robert, A. 2003. A model to assess relationships between forest dynamics and spatial structure. *Journal of Vegetation Science* 14: 823–834.
- Motta, R., Nola, P. & Piussi, P. 2002. Long-term investigations in a strict forest reserve in the eastern Italian Alps: spatio-temporal origin and development in two multi-layered subalpine stands. *Journal of Ecology* 90: 495–507.
- Murrell, D.J. 2009. On the emergent spatial structure of size-structured populations: when does self-thinning lead to a reduction in clustering? *Journal of Ecology* 97: 256–266.
- Nathan, R., Safriel, U., Noy-Meir, I. & Schiller, G. 2000. Spatiotemporal variation in seed dispersal and recruitment near and far from *Pinus halepensis* trees. *Ecology* 81: 2156–2169.
- Neuhäuslová, Z. (ed.) 2001. *Map of potential natural vegetation of the Šumava National Park. explanatory text. Silva Gabreta, Suppl. 1/2001*. Správa Národního parku Šumava, Vimperk, CZ.
- Niklasson, M. 2002. A comparison of three age determination methods for suppressed Norway spruce: implications for age structure analysis. *Forest Ecology and Management* 161: 279–288.
- Perry, G.L.W., Miller, B.P. & Enright, N.J. 2006. A comparison of methods for the statistical analysis of spatial point patterns in plant ecology. *Plant Ecology* 187: 59–82.
- Petrůš, J. & Neuhäuslová, Z. 2001. Pedology. In: Neuhäuslová, Z. (ed.) *Map of potential natural vegetation of the Šumava National Park. explanatory text, Silva Gabreta, Suppl. 1/2001*, pp. 84–85. Správa Národního parku Šumava, Vimperk, CZ.
- Plotkin, J.B., Chave, J.M. & Ashton, P.S. 2002. Cluster analysis of spatial patterns in Malaysian tree species. *The American Naturalist* 160: 629–644.
- Pomeroy, J., Marks, D. & Link, T. 2009. The impact of coniferous forest temperature on incoming longwave radiation to melting snow. *Hydrological Processes* 23: 2513–2525.
- Pommerening, A. & Stoyan, D. 2006. Edge-correction needs in estimating indices of spatial forest structure. *Canadian Journal of Forest Research* 36: 1723–1739.
- Seiler, J. & Matzner, E. 1995. Spatial variability of throughfall chemistry and selected soil properties as influenced by stem distance in a mature Norway spruce (*Picea abies*, Karst.) stand. *Plant and Soil* 176: 139–147.
- Sofron, J., Neuhäuslová, Z. & Wild, J. 2001. Climate. In: Neuhäuslová, Z. (ed.) *Map of potential natural vegetation of the Šumava National Park. explanatory text, Silva Gabreta, Suppl. 1/2001*, pp. 85–88. Správa Národního parku Šumava, Vimperk, CZ.
- Stoll, P. & Bergius, E. 2005. Pattern and process: competition causes regular spacing of individuals within plant populations. *Journal of Ecology* 93: 395–403.
- Stoyan, D. & Penttinen, A. 2000. Recent applications of point process methods in forestry statistics. *Statistical Science* 15: 61–78.
- Stoyan, D. & Stoyan, H. 1994. *Fractals, random shapes and point fields: methods of geometrical statistics*. John Wiley & Sons, Chichester, UK.
- Sturm, M. 1992. Snow distribution and heat flow in the taiga. *Arctic and Alpine Research* 24: 145–152.
- Svoboda, M., Fraver, S., Janda, P., Bače, R. & Zenáhlíková, J. 2010. Natural development and regeneration of a Central European montane spruce forest. *Forest Ecology and Management* 260: 707–714.

- Svoboda, M., Janda, P., Nagel, T.A., Fraver, S., Rejzek, J. & Bače, R. 2012. Disturbance history of an old-growth sub-alpine *Picea abies* stand in the Bohemian Forest, Czech Republic. *Journal of Vegetation Science* 23: 86–97.
- Swanson, M.E., Franklin, J.F., Beschta, R.L., Crisafulli, C.M., DellaSala, D.A., Hutto, R.L., Lindenmayer, D.B. & Swanson, F.J. 2011. The forgotten stage of forest succession: early-successional ecosystems on forest sites. *Frontiers in Ecology and the Environment* 9: 117–125.
- Takahashi, M., Sakai, Y., Ootomo, R. & Shiozaki, M. 2000. Establishment of tree seedlings and water-soluble nutrients in coarse woody debris in an old-growth *Picea abies* forest in Hokkaido, northern Japan. *Canadian Journal of Forest Research* 30: 1148–1155.
- Thomas, M. 1949. A generalization of Poisson's binomial limit for use in ecology. *Biometrika* 36: 18.
- Turkington, R. & Harper, J.L. 1979. The growth, distribution and neighbour relationships of *Trifolium repens* in a permanent pasture: I. Ordination, pattern and contact. *Journal of Ecology* 67: 201–218.
- Turner, M. 2010. Disturbance and landscape dynamics in a changing world. *Ecology* 91: 2833–2849.
- Watt, A.S. 1947. Pattern and process in the plant community. *Journal of Ecology* 35: 1–22.
- Wiegand, T. & Moloney, K.A. 2004. Rings, circles, and null-models for point pattern analysis in ecology. *Oikos* 104: 209–229.
- Wiegand, T., Gunatilleke, S., Gunatilleke, N. & Okuda, T. 2007. Analyzing the spatial structure of a Sri Lankan tree species with multiple scales of clustering. *Ecology* 88: 3088–3102.
- Wiegand, T., Martinez, I. & Huth, A. 2009. Recruitment in tropical tree species: revealing complex spatial patterns. *The American Naturalist* 174: E106–E140.
- Wild, J. 2001. *Prostorová mozaika bylinného patra šumavských smrčů z hlediska kůrovcové kalamity*. [Spatial pattern of herb layer vegetation in the mountain spruce forests in the Šumava Mts. influenced by bark beetle outbreak]. Aktuality Šumavského výzkumu, Srní 2.–4. dubna 2001, pp. 116–118. Správa NP a CHKO Šumava, Vimperk, CZ.
- Zenáhlíková, J., Svoboda, M. & Wild, J. 2011. Stav a vývoj přirozené obnovy před a jeden rok po odumření stromového patra v horském smrkovém lese na Trojmezí v Národním parku Šumava. [The state and development of natural regeneration before and one year after a dieback in the tree layer of a mountain]. *Silva Gabreta* 17: 37–54.
- Zielonka, T. 2006. Quantity and decay stages of coarse woody debris in old-growth subalpine spruce forests of the western Carpathians, Poland. *Canadian Journal of Forest Research* 36: 2614–2622.
- Zielonka, T., Holeksa, J., Fleischer, P. & Kapusta, P. 2010. A tree-ring reconstruction of wind disturbances in a forest of the Slovakian Tatra Mountains, Western Carpathians. *Journal of Vegetation Science* 21: 31–42.

Supporting Information

Additional Supporting Information may be found in the online version of this article:

Appendix S1. Numbers of recruits per plot with new or old snags as nearest neighbour.

Appendix S2. Comparison of pair correlation functions of spruce recruits and fitted Cox processes.

Appendix S3. Random and cluster components of recruit spatial pattern.